Module 6

Git Basics for Course Work

Learning objectives

Explain working tree, staging, and commits

Use status, add, commit, log, diff

Clone and push to GitHub

Write a useful .gitignore for build artifacts

Prerequisites

Module 5 (project structure)

git installed

Learning path

Learning Path

Diagram: Learning Path
(render with mmdc when available)

```
flowchart LR
A[Git model] --> B[init & clone]
B --> C[add & commit]
C --> D[remotes]
D --> E[gitignore]
E --> F[Module 7]
```

Course context

Course work is submitted via Git

Never commit build/ or large waveforms

Small focused commits are easier to review

Core topics

Self-check

`$ ./scripts/module6.sh --check`

Terminal: module_check

```
□[0;36m========================================□[0m
□[0;36mModule 6: Environment self-check□[0m
□[0;36m========================================□[0m

□[0;32m[OK]□[0m git available: git version 2.43.0
□[0;32m[OK]□[0m module6 directory exists: /mnt/d/proj/universal-verification-methodology/learn_unix_
□[0;32m[OK]□[0m gitignore/sample.gitignore exists

□[0;32m[OK]□[0m All required checks passed.
```

Basic workflow

```
git status
git add file.sv
git commit -m "Add counter testbench"
```

Git installation

```
$ git --version
```

Terminal: git_version

```
git version 2.43.0
```

Exploring git help

```
$ git help -a 2>/dev/null | head -5
```

Terminal: git_help

```
See 'git help <command>' to read about a specific subcommand

Main Porcelain Commands
   add                       Add file contents to the index
   am                        Apply a series of patches from a mailbox
```

Practice

Exercise scaffold

```
./scripts/module6.sh --scaffold
```

Exercises

Scaffold git_demo and make a commit

Write .gitignore for build/

git log --oneline

Summary & next steps

Local Git workflow

Remotes and GitHub

Next: Module 7 — collaboration

Next: Module 7: Advanced Git for Collaboration